

WEEK 2

Task1

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Source Code:

```
#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

#include <stdlib.h>

int main() {
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead;
    source = open("source.txt", O_RDONLY);
    if (source < 0) {
        perror("Error opening source file");
        return 1;
    }
    destination = open("destination.txt", O_WRONLY | O_CREAT | O_TRUNC,
0644);

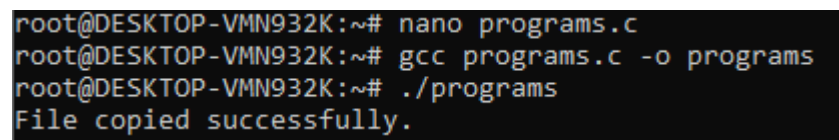
    if (destination < 0) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }
```

```

while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0) {
    write(destination, buffer, bytesRead);
}
close(source);
close(destination);
printf("File copied successfully.\n");
return 0;
}

```

Output:



```

root@DESKTOP-VMN932K:~# nano programs.c
root@DESKTOP-VMN932K:~# gcc programs.c -o programs
root@DESKTOP-VMN932K:~# ./programs
File copied successfully.

```

- The program starts execution in user space.
- The open () system call transfers control from user space to kernel space to open the file.
- After opening the file, control returns to user space with a file descriptor.
- The read () system call switches to kernel space to read data from the source file.
- The kernel copies the data to the user buffer and returns control to user space.
- The write () system call switches to kernel space to write data to the destination file.
- After writing the data, control returns to user space.
- The close () system call switches to kernel space to close the files and release resources.
- The kernel completes the operation and returns control to user space, where the program terminates.

Task2:

Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved

System Call	Purpose	Kernel Service
<code>execve()</code>	Executes the cat program	Process Management
<code>openat()</code>	Opens sample.txt	File System Management
<code>read()</code>	Reads data from the file	File I/O
<code>write()</code>	Displays data on the terminal	Output Management
<code>close()</code>	Closes the file	Resource Management
<code>exit_group()</code>	Terminates the process	Process Termination

Source code(sample.txt):

Hello Linux

Output:

```
root@DESKTOP-VMN932K:~# echo "Hello Linux" > sample.txt
root@DESKTOP-VMN932K:~# nano sample.txt
root@DESKTOP-VMN932K:~# cat sample.txt
Hello Linux
root@DESKTOP-VMN932K:~# strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7fff8fe918b8 /* 25 vars */) = 0
brk(NULL)                               = 0x59f300ffb000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x72305ecba000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19087, ...}) = 0
mmap(NULL, 19087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x72305ecb5000
close(3)                                = 0
```